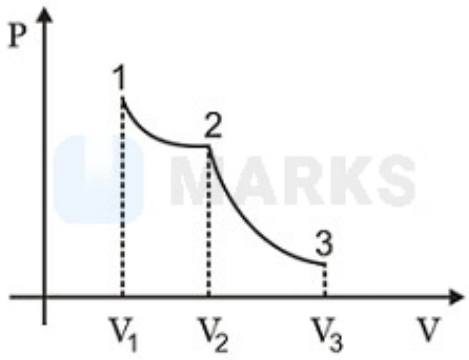
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

One mole of a perfect gas, initially at a pressure and temperature of 10^5 N m^{-2} and 300 K, respectively, expands isothermally until its volume is doubled and then adiabatically until its volume is again doubled. Find the final pressure of the gas. Given $\gamma = 1.4$.



(A) $(0.5)^{2.4} \times 10^5$ N m^{-2}

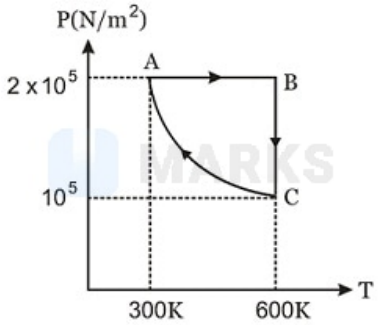
(B) $(0.5)^{1.4} \times 10^5$ N m^{-2}

(C) $(0.5)^{2.4} \times 10^6$ N m^{-2}

(D) $(0.5)^{1.4} \times 10^6$ N m^{-2}

Q.2

Two moles of a monatomic ideal gas are taken through a cyclic process shown on the P - T diagram. The process CA is represented as PT = constant. If the efficiency of the given cyclic process is $1 - \frac{x}{12 \ln 2 + 15}$, then what is the value of x?



(A) 23

(B) 21

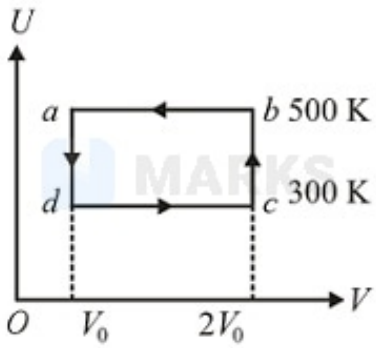
(C) 30

(D) 16

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Q.3

Diagram shows the variation in the internal energy U with the volume V of 2.0 moles of an ideal gas in cyclic process abcda. The temperatures of the gas at b and c are 500 K and 300 K, respectively. Calculate the heat absorbed by the gas during the process.



(A) 400 R ln 2

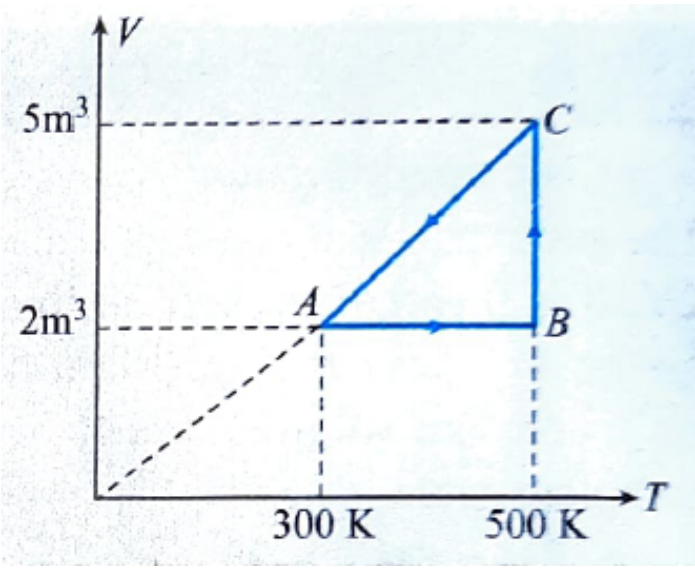
(B) 500 R ln 2

(C) 700 R ln 2

(D) 800 R ln 2

Q.4

The given figure shows a process $ABCA$ performed on 1mog of an ideal gas. Find the net heat supplied to the gas system during the process.



(A) 32.3 kJ

(B) 21.4 kJ

(C) 2.14 kJ

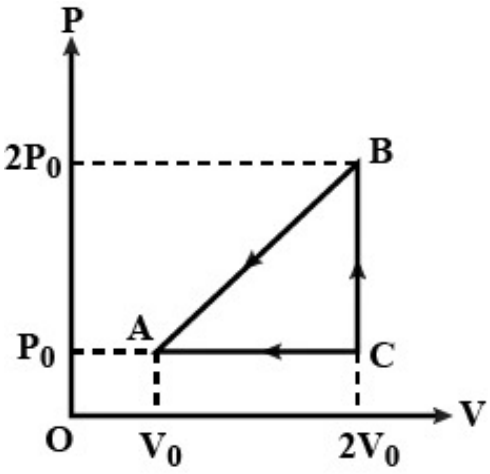
(D) 3.23 kJ

Q.5

One mole of a monatomic gas is taken from a point A to at another point B along the path ACB. The initial

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temperature A is T_0. Calculate the heat absorbed by the gas in the process $A \rightarrow C \rightarrow B$.



(A) $\frac{11RT_0}{2}$

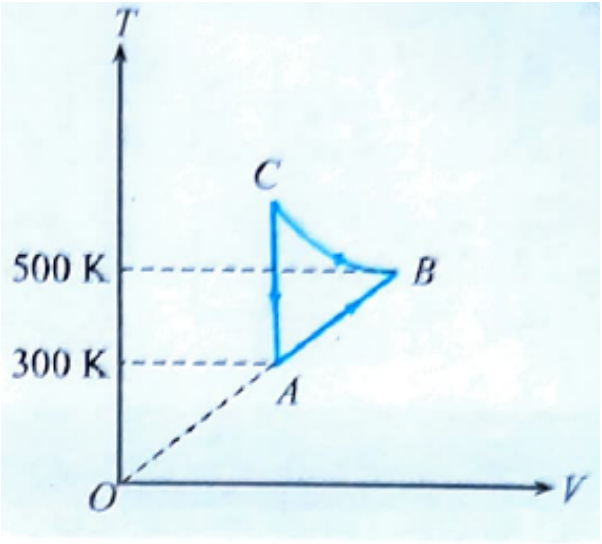
(B) $\frac{50RT_0}{2}$

(C) $\frac{300RT_0}{2}$

(D) $\frac{60RT_0}{2}$

Q.6

Consider the cyclic process $ABCA$, shown in the given figure performed, on a sample of 2.0 moles of an ideal gas. A total of $1200\ \text{J}$ of heat is withdrawn from the sample in the process. Find the work done by the gas during the part BC.



(A) -4000 J

(B) -5000 J

(C) -3507 J

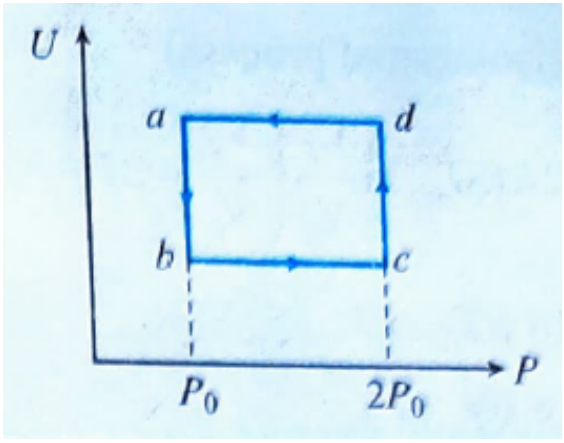
(D) -4524 J

Q.7

The given figure shows the variation of internal energy (U) with the pressure (P) of 2.0 mole gas in cyclic process $abcda$. The temperatures of gas at c and d are $300\ \text{K}$ and $500\ \text{K}$, respectively. Calculate the heat

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absorbed by the gas during the process.



(A) 400 R ln 2

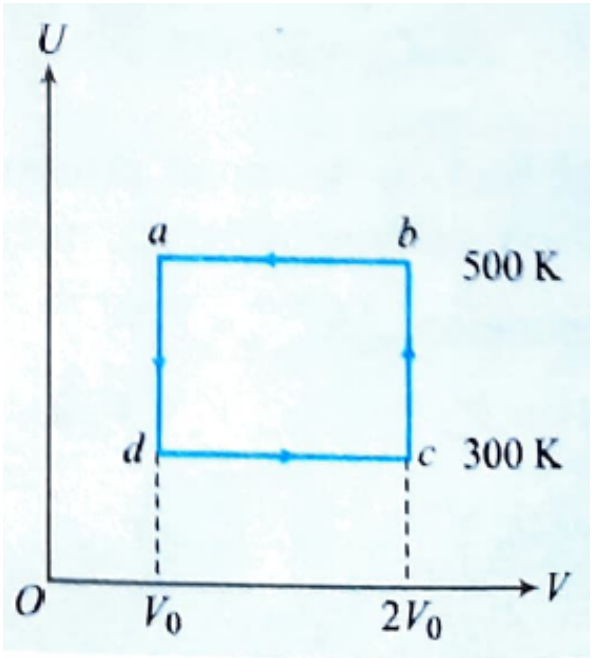
(B) 500 R ln 2

(C) 300 R ln 2

(D) 600 R ln 2

Q.8

The given figure shows the variation in the internal energy U with the volume V of 2.0 moles of an ideal gas in cyclic process $abcda$. The temperatures of the gas at b and c are 500 K and $300\ \mathrm{K}$, respectively. Calculate the heat absorbed by the gas during the process.



(A) 400 R ln2

(B) 500 R ln2

(C) 300 R ln2

(D) 600 R ln2

Q.9

Two moles of a certain ideal gas at a temperature 300 K were cooled isochorically so that the gas pressure is

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reduced $\eta\ (=2)$ times. The gas is expanded isobarically afterwards till its temperature got back to original value. The total amount of heat absorbed by gas in this process is $C_V = 1.5R$

(A) 2000 J

(B) 2500 J

(C) 3000 J

(D) 4000 J

Q.10

The temperature of a liquid does not increase during boiling. The heat energy supplied during this process,

(A) increases the kinetic energy of the molecules of the liquid

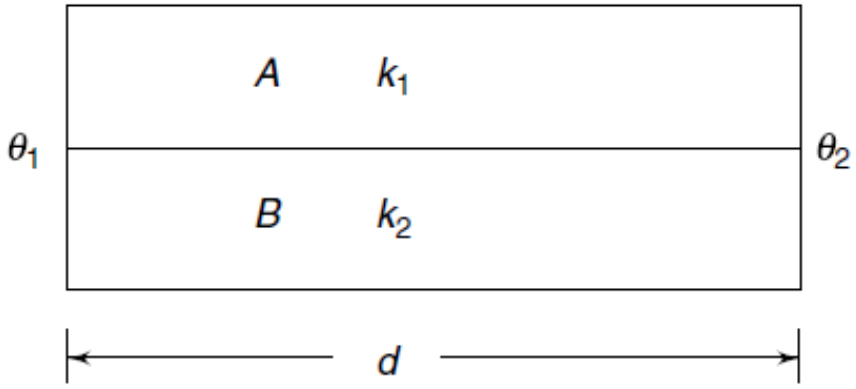
(B) increases the potential energy of the molecules

(C) increases both the kinetic and potential energy of the molecules

(D) is merely wasted since no increase occurs in the total energy of the molecules.

Q.11

Two rods A and B of different materials are welded together as shown in Fig. 8.10. If their thermal conductivities are k_1 and k_2, the thermal conductivity of the composite rod will be



(A) $2(k_1 + k_2)$

(B) $\frac{3}{2}(k_1 + k_2)$

(C) $(k_1 + k_2)$

(D) $\frac{1}{2}(k_1 + k_2)$

Q.12

A cylinder of radius R made of a material of thermal conductivity k, is surrounded by a cylindrical shell of inner radius R and outer radius $2R$ made of a material of thermal conductivity k_2. The two ends of the combined system are maintained at two different temperatures. There is no loss of heat across the cylindrical surface and the system is in steady state. The effective thermal conductivity of the system is

(A) $k_1 + k_2$

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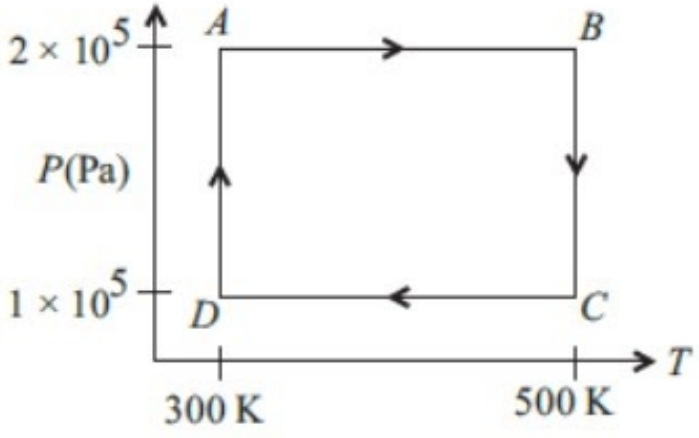
(B) $\frac{k_1 k_2}{k_1+k_2}$

(C) $\frac{k_1+3k_2}{4}$

(D) $\frac{3k_1+k_2}{4}$

Q.13

The work done on the gas in taking it from D to A is:



(A) $-690R$

(B) $+690R$

(C) $-414R$

(D) $+414R$

Q.14

The work of $146\ \text{kJ}$ is performed in order to compress one kilo mole of gas adiabatically and in this process the temperature of the gas increases by 7°C. The gas is a $\left(R = 8.3\ \text{J mol}^{-1}\ \text{K}^{-1}\right)$

(A) mixture of monoatomic and diatomic gases

(B) monoatomic gas

(C) diatomic gas

(D) triatomic gas

Q.15

Assuming the gas to be ideal the work done on the gas intaking it from A to B:(Consider two moles of helium gas)

(A) $400R$

(B) $500R$

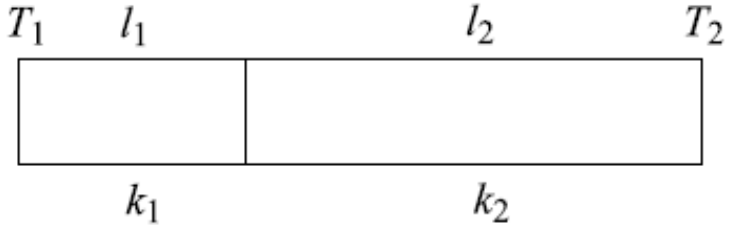
(C) $200R$

(D) $300R$

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Q.16

One end of a thermally insulated rod is kept at a temperature T_1 and the other at T_2. The rod is composed of two sections of lengths l_1 and l_2 and thermal conductivities k_1 and k_2, respectively. The temperature at the interface of the section is



(A) $(k_1 l_2 T_1 + k_2 l_1 T_2) / (k_1 l_1 + k_2 l_2)$

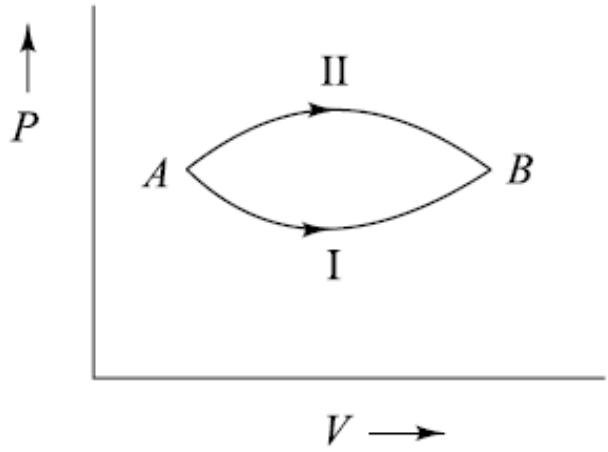
(B) $(k_2 l_1 T_1 + k_1 l_2 T_2) / (k_2 l_1 + k_1 l_2)$

(C) $(k_1 l_2 T_1 + k_2 l_1 T_2) / (k_1 l_2 + k_2 l_1)$

(D) $(k_1 l_1 T_1 + k_2 l_2 T_2) / (k_1 l_1 + k_2 l_2)$

Q.17

A gas is taken from state A to state B via two processes I and II as shown in the figure. If ΔU_1 and ΔU_2 are the changes in internal energy in the processes I and II respectively, then



(A) $\Delta U_1 = \Delta U_2$

(B) The relation between ΔU_1 and ΔU_2 cannot be determined

(C) $\Delta U_2 > \Delta U_1$

(D) $\Delta U_2 < \Delta U_1$

Q.18

Two moles of a monoatomic ideal gas occupy a volume V at $27^\circ\mathrm{C}$. The gas is expanded adiabatically to volume $2\sqrt{2}V$. Gas constant $R = 8.3\mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$.
The final temperature of the gas is

(A) $\frac{150}{\sqrt{2}}$ K

(B) 150 K

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(C) $150^{\circ}\mathrm{C}$

(D) $13.6^{\circ}\mathrm{C}$

Q.19

In an adiabatic process, the pressure P of a gas varies with temperature T as $P = kT^{5/2}$ where k is a constant. The ratio $\gamma\ (= C_p/C_v)$ of the gas is

(A) $\frac{7}{5}$

(B) $\frac{5}{3}$

(C) $\frac{3}{2}$

(D) $\frac{9}{5}$

Q.20

In an adiabatic process, the relation between internal energy U, volume V and pressure P of a real gas is $U = a + bPV$
where a and b are constants. If $a = 3$ joule and $b = 2.5$, the ratio of the two specific heats of the gas is

(A) $\frac{7}{5}$

(B) $\frac{5}{3}$

(C) $\frac{6}{5}$

(D) $\frac{3}{2}$

Answers & Solutions

Q.1 Answer:

$0.5^{2\cdot4} \times 10^5$ N m^{-2}

Solution:

Let P_1, V_1, T_1 be the initial pressure, volume and the temperature of the gas, respectively. For isothermal expansion (1 to 2), we have

$$V_2 = 2V_1 \quad ; \quad P_1V_1 = P_2\ V_2$$

$$P_2 = P_1\frac{V_1}{V_2} = \frac{P_1}{2} = 0.5 \times 10^5 \text{ N } m^{-2}$$

$$T_2 = T_1 = 300 \text{ K}$$

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For adiabatic expansion (2 to 3),

$$P_2V_2^{\gamma} = P_3V_3^{\gamma}$$

$$\Rightarrow \quad V_3 = 2V_2$$

$$\Rightarrow \quad P_3 = P_2\frac{V_2}{V_3}^{\gamma} = 0.5 \times 10^5 \times 0.5^{1.4}$$

$$\Rightarrow \quad P_3 = 0.5^{2.4} \times 10^5 \quad N \quad m^{-2}$$

Q.2 Answer:

21

Solution:

For process AB**,**

$$T_A = 300 \quad K, \quad T_B = 600 \quad K$$

$$W = nR\,\Delta\,T = nRT_B - T_A = 300nR = 600R$$

$$Q = nC_p\,\Delta\,T = 2 \times \frac{5}{2}R300 = 1500R$$

For process BC**,**

$$W = nRT\ln\frac{V_f}{V_i} = nRT\ln\frac{p_i}{p_f}$$

$$= nRT\ln 2 = 1200R\ln 2$$

For process CA**,**

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$$W = \int P\,dV = \int_{600}^{300} \frac{K}{T}\frac{2nRT}{K}dT$$

$$= -2nR300 = -1200R$$

$$Q = nC_v \Delta T + W$$

$$= 2 \times \frac{3}{2}R - 300 - 1200R$$

$$= -900R - 1200R = -2100R$$

$$\eta = \frac{600R + 1200R \ln 2 - 1200R}{1500R + 1200R \ln 2}$$

$$\eta = 1 - \frac{21}{12 \ln 2 + 15}$$

Comparing with given equation, $x = 21$

Q.3 Answer:

400 R ln 2

Solution:

In the process a to b and c to d

as $\Delta U = 0$**, therefore** $\Delta T = 0$ **or T = constant**

$$W = \int_{V_i}^{V_f} PdV$$

We have,

$$PV = nRT \Rightarrow P = \frac{nRT}{V}$$

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$$\therefore \quad W = \int_{V_i}^{V_f} nRT\frac{RV}{V}$$

$$= nRT\ln V_{V_i}^{V_f} = nRT \ln \frac{V_f}{V_i}$$

$$W_{ab} = nRT_b \ln \frac{2V_0}{V_0} = 2R \times 500 \ln 2 = 1000\ R \ln 2$$

and $W_{cd} = nRT_c \ln \frac{V_0}{2V_0} = 2R \times 300 \ln \frac{1}{2} = -600\ R \ln 2$

There is no volume change from b to c and from d to a, so

$$W_{bc} = W_{da} = 0$$

The work done in the complete cycle

$$W = W_{ab} + W_{bc} + W_{cd} + W_{da}$$

$$= 1000\ R \ln 2 + 0 - 600\ R \ln 2 + 0$$

$$= 400\ R \ln 2$$

Q.4 Answer:

2.14 kJ

Solution:

Sai. As we know in a cyclic process gas finally returns to its initial state and total change in internal energy of gas is zero thus the total heat supplied to the gas is equal to work done the gas. Now we find the work done by the gas in different put of the cycle.

In process AB : As shown in graph, during process AB, volume of gas remains constant; thus work done by gas is zero.

$W_{AB} = 0$

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In Process BC : In process BC, volume of gas changes from $V_1 = 2\ \mathrm{m}^3$ to $V_2 = 5\ \mathrm{m}^3$ thus work done can be obtained as

$W_{BC} = \int_2^5 pdV$

As in process BC, temperature of gas remains constant at 5ω. we can write pressure of gas from gas law as

$P = \frac{RT}{V} = \frac{500R}{V}$

Now work done is

$\therefore \quad W_{BC} = \int_2^5 \frac{500R}{V} dV = 500R \ln\left(\frac{5}{2}\right)$

In process CA

As in this process, path is a straight line passing through origin thus $V \propto T$ or pressure of gas remains constant and we know its gas pressure is constant work done is given as

$W_{CA} = nR\left(T_2 - T_1\right) = nR(300 - 500) = -200R$

This negative as gas is being compressed from volume $5\ \mathrm{m}^3$ to $2\ \mathrm{m}^3$ work is done on the gas.

Now we can find the total work done by the gas in the complete cycle ABCA as

$W_{ABCA} = W_{AB} + W_{BC} + W_{CA}$

$= -0 + 500R \ln\left(\frac{5}{2}\right) - 200R - R\left(500 \ln \frac{5}{2} - 200\right)$

$= 2146.22\ \mathrm{J} =$ **heat supplied to the gas**

Q.5 Answer:

$\frac{11RT_0}{2}$

Solution:

If T_B be the temperature at B, then by gas law

$\frac{P_A V_A}{T_A} = \frac{P_B V_B}{T_B}$

$\therefore \quad T_B = \frac{P_B V_B}{P_A V_A} T_A = \frac{(2P_0)(2V_0)}{P_0 V_0} T_0$

The change in internal energy from A to B

$\Delta U = nC_v \Delta T = 1 \times \frac{3R}{2} \times (4T_0 - T_0) = \frac{9RT_0}{2}$

Work done in the process A to C

$W_{AC} = P\Delta V = P_0\left(2V_0 - V_0\right) = P_0 V_0 = RT_0$ and $W_{CB} = 0$

$\therefore$ Total work done from $A \to C \to B$

$W_{AC} + W_{CB} = RT_0 + 0 = RT_0$

From the first law of thermodynamics, $Q = \Delta U + W$

$= \frac{9RT_0}{2} + RT_0 = \frac{11RT_0}{2}$

Thus heat absorbed by the gas from $A \to C \to B$ is $\frac{11RT_0}{2}$

Q.6 Answer:

-4524 J

Solution:

In a cyclic process $\Delta U = 0$

From the first law of thermodynamics, for the cyclic process

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$Q = \Delta U + W$

$\therefore \quad W = Q - \Delta U = -1200 - 0 = -1200 \text{ J}$

From C to A, $\Delta V = 0$

$\therefore \quad W_{CA} = 0$

For the whole cycle

$W_{AB} + W_{BC} + W_{CA} = W = -1200 \text{ J}$ **...(i)**

As $W_{CA} = 0$

$\therefore \quad W_{AB} + W_{BC} = -1200 \text{ J}$

Work done from A to B : In the process $V \propto T$, so pressure remains constant.

We know that $PV = nRT$

or $P\Delta V = nR\Delta T$

$\therefore \quad W_{AB} = P\Delta V = nR\Delta T = 2 \times 8.31 \times (500 - 300) = 3324 \text{ J}$

Substituting this value in Eq. (i), we get $3324 + W_{BC} = -1200$

$\therefore \quad W_{BC} = -4524 \text{ J}$

Q.7 Answer:

400 R ln 2

Solution:

Change in internal energy for cyclic process $(\Delta U) = 0$ **For process** $a \to b$, $(P-$ **constant)**

$W_{a\to b} = P\Delta V = nR\Delta T = -400R$

For process $b \to c (T$ **- constant** $)$ $W_{b\to c} = -2R(300)\ln 2$

For process $c \to d (P-$ **constant** $)$ $W_{c\to d} = +400R$

For process $d \to a (T$ **- constant** $)$ $W_{d\to a} = +2R(500)\ln 2$

Net work, $(\Delta W) = W_{a\to b} + W_{b\to c} + W_{c\to d} + W_{d\to a}$

$\Delta W = 400R\ln 2$

$\Delta Q = \Delta U + \Delta W$**, first law of thermodynamics.**

$\Delta Q = 400R\ln 2$

Q.8 Answer:

400 R ln2

Solution:

In the process a to b and c to d as $\Delta U = 0$, therefore $\Delta T = 0$ or $T =$ constant

$W = \int_{V_i}^{V_f} PdV$

We have $pV = nRT \Rightarrow P = \frac{nRT}{V}$

$\therefore W = \int_{V_i}^{V_i} (nRT)\frac{RV}{V} = nRT|\ln V|_{V_i}^{V_f} = nRT\ln\frac{V_f}{V_i}$

$\mathrm{W}_{ab} = nRT_b \ln\frac{2V_0}{V_0} = 2R \times 500\ln 2 = 1000R\ln 2$

and $\mathrm{W}_{cd} = nRT_c \ln\frac{V_0}{2V_0} = 2R \times 300\ln\frac{1}{2} = -600R\ln 2$

volume change from b to c and from d to a, so in no

$\mathrm{W}_{bc} = \mathrm{W}_{da} = 0$

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work done in complete cycle

$W = W_{ab} = W_{bc} = W_{cd} = W_{da} = 0$

$= 1000R\ln 2 + 0 - 600R\ln 2 + 0 = 400R\ln 2$

Q.9 Answer:

2500 J

Solution:

For isochoric process, $P \propto T$

As pressure is halved,

so, the new temperature is $\frac{300}{2} = 150 \quad K$

$\Delta Q_{V=constant} = \mu C_V \Delta T$

$= 2 \times 1.5R \times (-150) = -450R \qquad ...(1)$

From Mayor's relation, $C_P = C_V + R = 2.5R$

$\Delta Q_{P=constant} = 2 \times 2.5R \times 150 = 750R \qquad ...(2)$

Total heat absorbed

$Q = \Delta Q_{P=constant} + \Delta Q_{V=constant}$

$\Rightarrow Q = 750 - 450R = 300 \times 8.3$

$\Rightarrow Q = 2490\ J \approx 2500\ J$

Q.10 Answer:

increases the potential energy of the molecules

Solution:

increases the potential energy of the molecules

Q.11 Answer:

$\frac{1}{2}(k_1 + k_2)$

Solution:

If the ends of the rods are maintained at temperatures θ_1 **and** θ_2**, the rate of flow of heat through each rod is given by**

$Q_1 = \frac{k_1 A(\theta_1 - \theta_2)}{d}$

and $Q_2 = \frac{k_2 A(\theta_1 - \theta_2)}{d}$

where A **is the cross-sectional area of each rod and** d **is the length of each rod. Now the cross-sectional area of the composite rod is** $2A$**. If** k **is the thermal conductivity of the composite rod, the**

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rate of heat flow is

$Q = \frac{k(2A)(\theta_1-\theta_2)}{d}$

But $Q = Q_1 + Q_2$**. Therefore**

$\frac{k(2A)(\theta_1-\theta_2)}{d} = \frac{k_1A(\theta_1-\theta_2)}{d} + \frac{k_2A(\theta_1-\theta_2)}{d}$

Or $k = \frac{1}{2}(k_1 + k_2)$

Hence the correct choice is (d).

Q.12 Answer:

$\frac{k_1+3k_2}{4}$

Solution:

Let the length of the cylinder be l **and let its ends be maintained at temperatures** θ_1 **and** θ_2**. Area of the cross-section of the inner cylinder** $= \pi R^2$**. Area of cross-section of outer cylinder** $= \pi(2R)^2 - \pi R^2 = 3\pi R^2$**.**

Rate of flow of heat across inner cylinder is

$Q_1 = \frac{k_1\pi R^2(\theta_1-\theta_2)}{l}$ **(i)**

Rate of flow of heat across the outer shell is

$Q_2 = \frac{k_2\left(3\pi R^2\right)(\theta_1-\theta_2)}{l}$ **(ii)**

Let the effective thermal conductivity of the compound cylinder be k**. The rate of flow of heat across the compound cylinder is**

$Q = \frac{k\left(4\pi R^2\right)(\theta_1-\theta_2)}{l}$ **(iii)**

Now $Q = Q_1 + Q_2$ **(iv)**

Using (i), (ii) and (iii) in (iv), we get $4k = k_1 + 3k_2$

Or $k = \frac{k_1+3k_2}{4}$

Hence the correct choice is (c).

Q.13 Answer:

$+414R$

Solution:

Process D **to** A **is isothermal. Since the pressure increases from** D **to** A**, the volume of the gas decreases. hence work is done on the gas. Thus,** $W_{D\to A}$ **is negative**

$$W_{D\to A} = 2.303nRT\log 10\left(\frac{P_D}{P_A}\right)$$

$$= -2.303 \times 2 \times R \times 300\log 10\left(\frac{1\times 10^5}{2\times 10^5}\right)$$

$$= +414\text{R}$$

Q.14 Answer:

diatomic gas

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Solution:

An adiabatic process, the work is given by
$W = \frac{nR(T_1 - T_2)}{\gamma - 1}$
or $146 \times 10^3 = \frac{1000 \times 8.3 \times 7}{(\gamma - 1)} \quad (\because \Delta T = 7^\circ\text{C} = 7\text{ K})$
which gives $\gamma = 1 + 0.4 = 1.4$. **Hence the gas is diatomic**

Q.15 Answer:

$400R$

Solution:

$PV = nRT$. **Since** $P =$ **constant in process** A **to** B $W_{A \to B} = P\Delta V = nR\Delta T = 2 \times R \times (500 - 300) = 400R$

Q.16 Answer:

$(k_1 l_2 T_1 + k_2 l_1 T_2) / (k_1 l_2 + k_2 l_1)$

Solution:

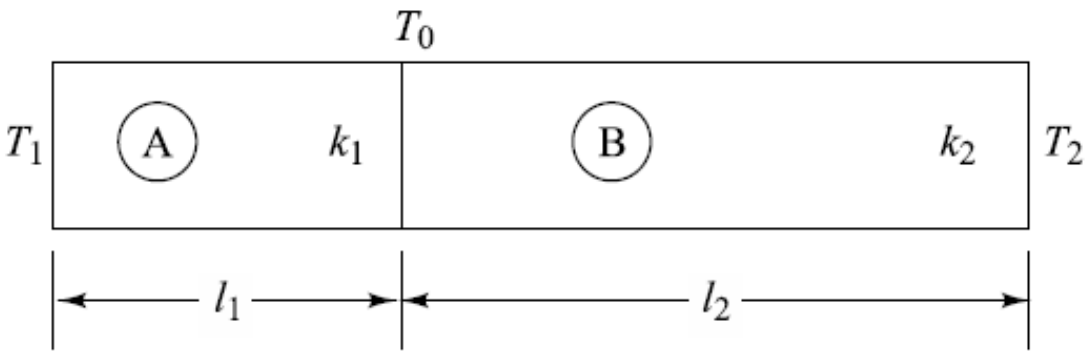
Let T_0 **be the temperature at the interface and** A **be the cross-sectional area of each rod.**
In the steady state, the rate of flow of heat through $\text{rod } A =$ **rate of flow of heat through** $\text{rod } B$**, i.e.**

$$\frac{Q_1}{t} = \frac{Q_2}{t}$$

$$\Rightarrow \frac{k_1 A (T_1 - T_0)}{l_1} = \frac{k_2 A (T_0 - T_2)}{l_2}$$

$$\Rightarrow k_1 l_2 (T_1 - T_0) = k_2 l_1 (T_0 - T_2)$$

$$\Rightarrow T_0 = \frac{(k_1 l_2 T_1 + k_2 l_1 T_2)}{(k_1 l_2 + k_2 l_1)}$$



Q.17 Answer:

$\Delta U_1 = \Delta U_2$

Solution:

The change in internal energy is independent of the path of followed by the processes; it depends only on the initial and the final states of the system. Hence $\Delta U_1 = \Delta U_2$

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Q.18 Answer:

150 K

Solution:

$T_1 = 300\ \text{K}, V_1 = V, V_2 = 2\sqrt{2}V$

Let T_2 be the final temperature of the gas. T_2 is obtained from the adiabatic relation

$$T_1 V_1^{(\gamma-1)} = T_2 V_2 (\gamma - 1)$$

$$\text{Or } T_2 = T_1 \times \left(\frac{V_1}{V_2}\right)^{(\gamma-1)}$$

For a monoatomic gas $\gamma = \frac{5}{3}$. Therefore,

$$T_2 = 300 \times \left(\frac{1}{2\sqrt{2}}\right)^{2/3} = 150\ \text{K}$$

So, the correct choice is (b).

Q.19 Answer:

$\frac{5}{3}$

Solution:

For an adiabatic process, $P^{1-\gamma}T\gamma =$ constant

or $\frac{T^\gamma}{P^{1-\gamma}} =$ constant

or $p^{1-\gamma} = \frac{T^\gamma}{\text{constant}}$

or $P \propto T^{\frac{\gamma}{\gamma-1}}$

Given $P \propto T^{5/2}$

$$\therefore \quad \frac{\gamma}{\gamma-1} = \frac{5}{2} \quad \Rightarrow \quad \gamma = \frac{5}{3}$$

So the correct choice is (b).

Q.20 Answer:

$\frac{7}{5}$

Solution:

From the first law, $dQ = dU + PdV$. For an adiabatic process $dQ = 0$. Therefore,

$$dU + PdV = 0$$

$$\Rightarrow d(a + bPV) + PdV = 0$$

$$\Rightarrow 0 + bPdV + bVdP + PdV = 0$$

$$\Rightarrow \quad (b+1)PdV + bVdP = 0$$

Dividing by PV,

$$(b+1)\frac{dV}{V} + \frac{bdP}{P} = 0$$

Integrating,

$$(b+1)\log V + b\log P = \text{constant}$$

$$\Rightarrow V^{b+1}P^b = \text{ constant}$$

$$\Rightarrow PV^{\frac{b+1}{b}} = \text{ constant}$$

For an adiabatic process $PV^\gamma =$ constant. Hence

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$\gamma = \frac{b+1}{b} = \frac{2.5+1}{2.5} = \frac{7}{5}$

So the correct choice is (a).

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